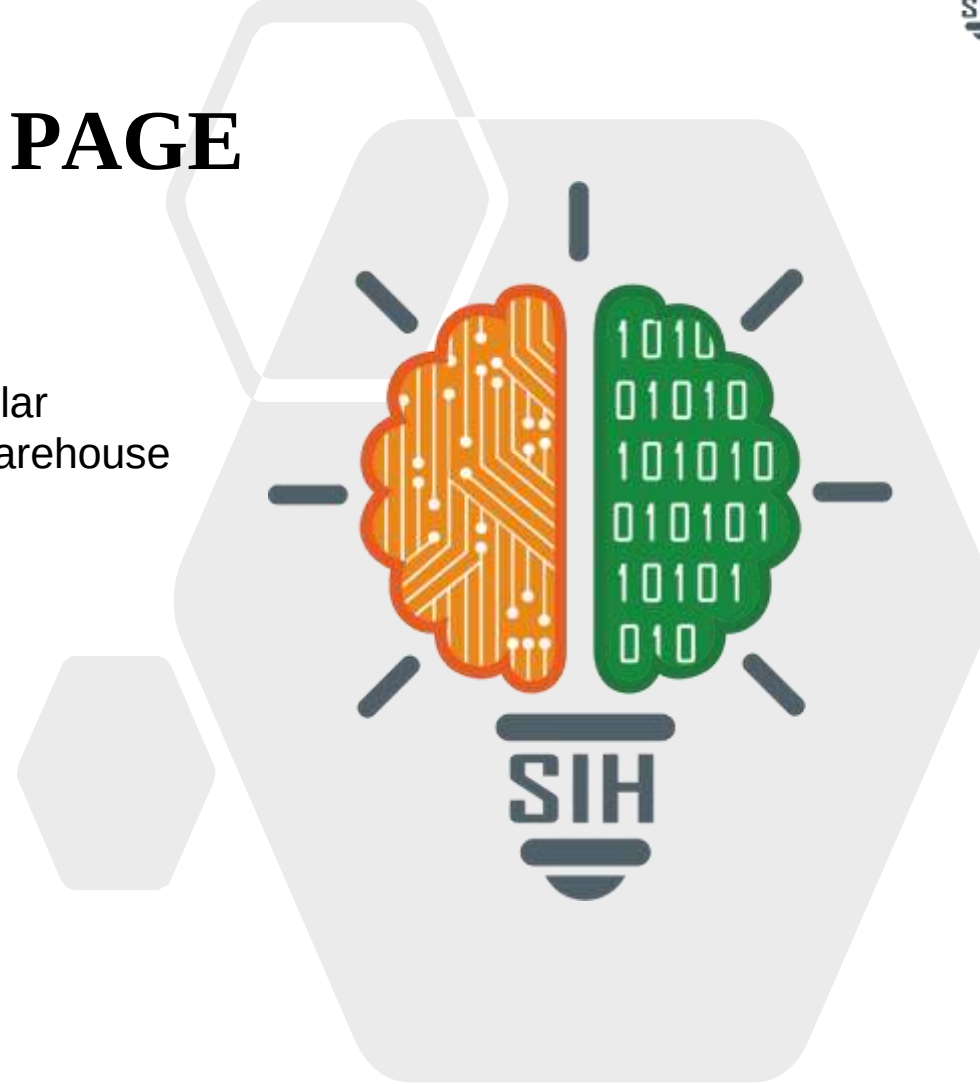


TITLE PAGE

- **Problem Statement ID** - SIH26112
- **Problem Statement Title** - Design and Develop a Modular Autonomous Mobile Robot (AMR) Platform for Smart Warehouse Automation
- **Theme** - Robotics and Drones
- **PS Category** - Hardware
- **Team ID** - 158819
- **Team Name (Registered on portal)** – Team_7



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

The Problem:

- Conventional AMRs primarily transport items at floor level.
- Multi-level racks require a mechanism to access elevated storage locations.
- Fixed AS/RS solutions are typically tied to dedicated rack and aisle infrastructure.
- Different rack heights require flexible vertical positioning during retrieval.
- Existing warehouses need solutions that can integrate without major rack modification.

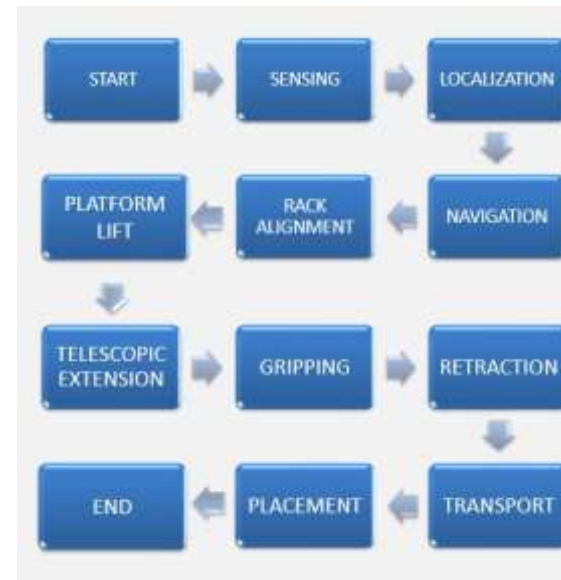
Our Solution:

- Autonomous robot for rack item retrieval
- Rising platform reaches the required rack level
- Telescopic arm extends into the rack
- Adjustable gripper accommodates different box sizes
- Designed for existing warehouse racks

Key Innovation:

- Mobile + lift + telescopic reach in one robot
- Automated rack-level access across different heights
- Adaptive gripping for varied item/box sizes
- Minimal rack modification for deployment
- Modular mechanical architecture for scalable development

How It Works:



TECHNICAL APPROACH

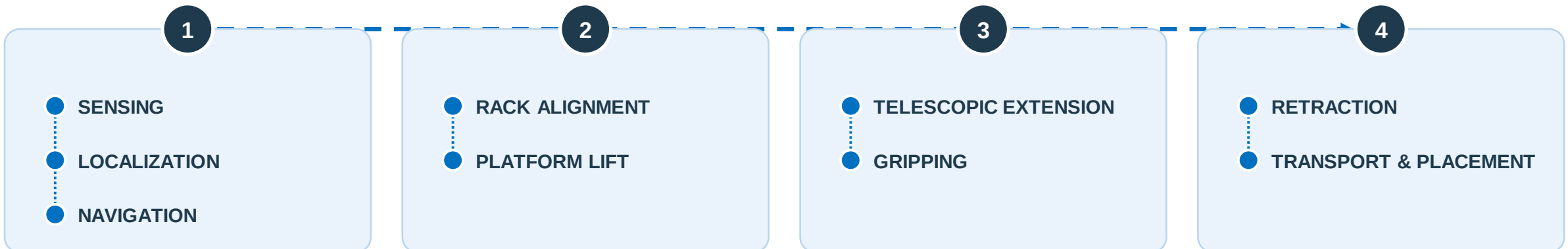
A. Hardware

AMR chassis	Adjustable rubber-padded gripper
Motor drivers	LiDAR / depth sensing
Battery system	IMU & wheel encoders
Camera	Proximity / safety sensors
Telescopic arm	Raspberry Pi 5 controller

B. Software

ROS 2	Rviz2
SLAM	Autodesk Fusion
Gazebo	Movelt 2
Nav2	Codesys

C. System Workflow



FEASIBILITY AND VIABILITY



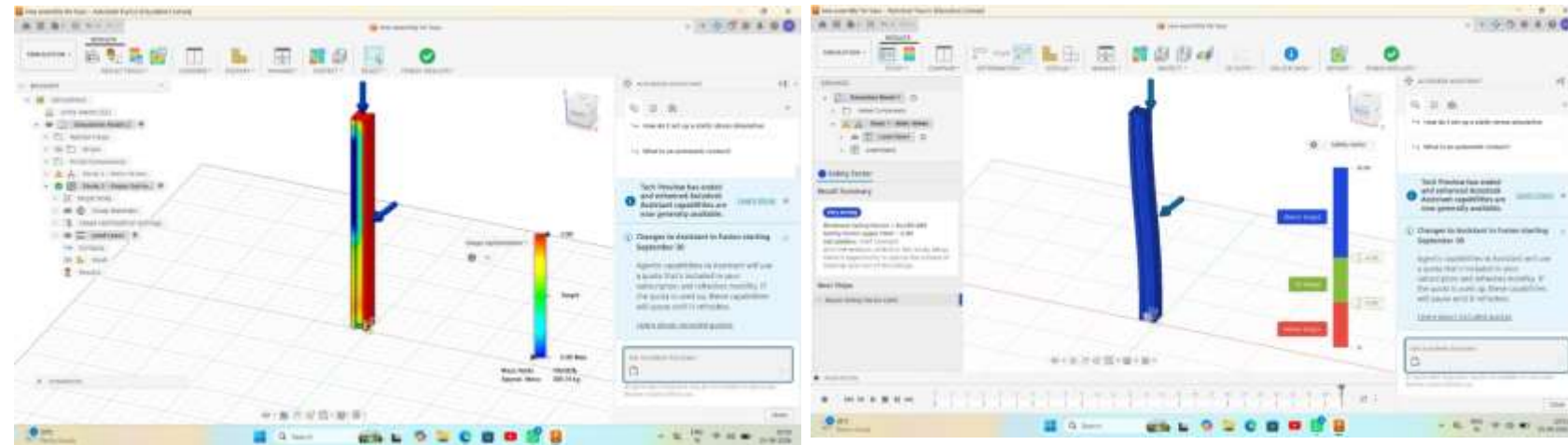
ASPECT	FEASIBILITY	KEY RISK	MITIGATION
Navigation & localization	ROS 2 + LiDAR navigation is well established	Dynamic obstacles, localization error	Sensor fusion + obstacle detection
Rack alignment	Depth / proximity sensing supports alignment	Lateral offset at the rack	Closed-loop fine alignment before lift
Lifting mechanism	Actuated linear lifting is readily implementable	Load variation, mechanical deflection	Payload limits + structural analysis + position feedback
Telescopic arm & gripper	Controlled linear reach with adjustable grip	Cantilever load, varied item sizes	Payload limits + support / locking + grip range limits
Stability & payload	Low-mounted battery supports stability	CG shift during arm extension	Wide base + low CG + extension limits
Safety	Sensors can support safe operation	Collision with people or racks	Obstacle sensing + speed limits + emergency stop
Manufacturing & maintenance	Modular design with standard components	Wear in moving parts, alignment drift	Modular parts + periodic calibration

Assembly file - <https://a360.co/3Vkj5IU>

For full details of the design with the topology details visit GitHub - <https://github.com/Yuvaraaj-007/SiH-26112/tree/main>

- **Warehouse Operations:** Automated multi-level retrieval; less repetitive manual handling
- **Productivity:** Continuous material movement; consistent retrieval workflow
- **Safety:** Less manual heavy handling; controlled robotic retrieval
- **Infrastructure:** Built for existing racks; minimal rack modification
- **Scalability:** Modular mechanical design; adapts to warehouse layouts
- **Economic Potential:** Potentially less manual handling; reusable multi-task platform

For assembly animation [click here](#)
For stress analysis video [click here](#)
For optimization of design [click here](#)



RESEARCH AND REFERENCES



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